MIL-F-37987A
14 August 1987
SUPERSEDING
MIL-F-37987 (DLA-DM)
3 JULY 1986

MILITARY SPECIFICATION

FORCEPS, VASCULAR TISSUE (DeBakey)

This specification is approved use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a straight vascular tissue forceps of the DeBakey design.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 S. 20th Street, Philadelphia, PA 19101, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Set-up.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-L-10547	- Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads
MIL-STD-794	- Parts and Equipment, Procedures for Packaging and Packing of

2.1.2 Drawing. The following drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

DRAWING

Defense Personnel Support Center (DPSC)

23362 - FORCEPS, VASCULAR TISSUE

(Copies of specifications and standards, and drawing required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.

E 92 - Vickers Hardness of Metallic Materials, Tests for.

E 140 - Hardness Conversion Tables for Metals (Relationship Between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness and Knoop Hardness).

F 899 - Stainless Steel Billet, Bar and Wire for Surgical Instruments.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The vascular tissue forceps shall be in accordance with the applicable requirements of GG-I-526, and as specified herein.

3.2 Material. The forceps shall be fabricated from Class 4, martensitic stainless steel in accordance with ASTM F899.

3.3 Hardness. After proper stress relieving, the forceps shall have a Rockwell hardness shall be C38 to C44. The requirements of GG-I-526 shall apply.

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3.4 Style, design and dimensions. Shall be a straight, vascular tissue forceps of the DeBakey design used for thoracic and vascular surgery. The forceps shall be approximately 12 inches long with a narrow distal end for handling small blood vessels or suture material. The serrated portion of the jaws shall be approximately 1 inch long and mesh together at the distal tip. In addition to lateral serrations, the jaws shall have 1 x 2 longitudinal serrations for alignment purposes. The handle portion of the forceps shall have lateral grooves or similar configuration for a sure grip. The forceps shall be in accordance with Defense Personnel Support Center (DPSC) Drawing 23362 for the appropriate type and National Stock Number (NSN).

3.5 Finish. Finish shall be non-glare, satin or matte. The matte finish shall exhibit a uniform nondirectional surface texture equivalent to that produced by lightly blasting the prepared surface with fine abrasives under pressure.

3.5.1 Passivation. The forceps shall be properly passivated.

3.5.2 Corrosion resistance. The forceps shall show no signs of corrosion or plating of copper when tested as specified in GG-I-526.

3.6 Identification markings. In addition to the requirements specified in GG-I-526, forceps of foreign manufacture shall be marked "stainless".

3.7 Workmanship. The forceps shall be free from burrs, cracks, pits, sharp edges, scale, corrosion, or other defects which detract from its appearance or impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements in sections 3 and 5. The inspection set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection. Inspection shall be in accordance with GG-I-526.

4.1.3 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.4 Certificates of quality. Certificates of quality supplied by the manufacturer of the corrosion-resistant steel may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the item identification, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in table I. Unit of product for examination purpose shall be one forceps.

TABLE I. Sampling criteria for inspection

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test purpose shall be one forceps. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling criteria for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Hardness test	3.3 & GG-I-526	4.4.1	S-2	1.5
Dimensions	3.4	---	S-2	2.5
Copper sulfate test	GG-I-526	GG-I-526	S-2	1.0
Boil test	GG-I-526	GG-I-526	S-2	1.0

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in the manufacture of the forceps shall be selected for chemical analysis.

4.3 Examination.

4.3.1 Examination of forceps. The forceps shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any forceps in the sample containing one or more defects shall be rejected and if the number of defective forceps in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

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TABLE III. Classification of defects

Categories	Defects
Critical	None defined.
Major	
101	Forceps not of the specified design for appropriate type, stock number and item identification.
102	Jaw serrations not of uniform depth and not well defined.
103	Jaw serrations not chamfered throughout full length.
104	Forceps not free from cracks, pits, burs, sharp edges, scale, rough areas or corrosion.
105	Finish of forceps not uniform and not as specified.
106	Jaws possess sharp tips.
Minor	
201	Forceps not free from dirt, polishing material or foreign matter.
202	Identification markings not present, not correct, not complete, or not legible.

4.3.2 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Hardness test. Hardness shall be determined in accordance with ASTM Standards E 18 or E 92. Conversion of Vickers hardness values to Rockwell values shall be in accordance with ASTM Standard E 140.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

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5.1.1 Level A.

5.1.1.1 Cleaning and drying. Forceps shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Each forceps shall be wrapped in neutral tissue paper or other suitable anticorrosive material. Each forceps shall be further packaged in accordance with MIL-P-116, submethod IC-1. Closure shall be as specified in the applicable box specification

5.1.1.4 Intermediate package. Twelve forceps shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, Type CF, class domestic. Closure of box shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for forceps shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Three hundred and sixty (360) forceps shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B-636, class weather resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Three hundred and sixty (360) forceps shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1 or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for forceps shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in a exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

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5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination exceed 250 pounds or 20 cubic feet. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 20 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 54 inches in height, 48 inches in width, and 40 inches in length. Pallet shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. The packaged forceps packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.2.5.2 Level C. The packaged forceps packed as specified in 5.2.3 shall be unitized as specified in 5.2.5.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container, and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.3.2.

6. NOTES

6.1 Intended use. The forceps shall be used for thoracic and vascular surgery.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Applicable National Stock Number.

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6.3 National Stock Number coverage. This specification covers the following National Stock Number:

National Stock No.	Item Identification
6515-01-139-8192	FORCEPS, VASCULAR TISSUE, DeBAKEY, Straight, 12 Inch (30.5 cm)

6.4 Subject term (key word) listing.

DeBakey forceps
Forceps
Vascular tissue forceps

6.5 Matte and satin finish. It should be noted that matte finish specified herein can be distinguishable from mirror and satin finishes under high-power illuminated magnification. Forceps of matte finish shall have a reduced reflective surface that is more glare resistant than satin finish.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-3863

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-37987A	FORCEPS, VASCULAR TISSUE (DeBakey)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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